Failure budgets and service objectives

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make the denominator and historical record part of the reliability claim.

Teach overlap-aware counting,correct refusals,integer boundaries and hypothetical versus actual retry evidence.

Keep one habit: Model the interval calculation → inspect counting probes → preserve original/counterfactual ledgers → assess a fresh objective → plan continued monitoring.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of evaluation pipelines

Reference example, not the only acceptable wording: Teacher explanation: Evaluation pipelines preserve case versions, procedure versions and raw outcomes before producing metrics. A rerun must identify what changed.

Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing model-dependent behaviour

Reference example, not the only acceptable wording: Teacher explanation: Model-dependent behaviour may vary between runs.

Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions.

Repeats estimate observed variation for this case; they are not additional independent users.

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of authentication and authorisation

Reference example, not the only acceptable wording: Teacher explanation: Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource.

Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Pictured end goal

Explain **failure budget**s and **service objective**s using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Evaluation pipelines

Testing model-dependent behaviour

Authentication and authorisation

Use the three-session teaching cycle above, with extra time as needed. **Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach objective scope,exact counts and preserved histories

> Separate aggregate reliability accounting from severe-incident and future-performance claims.

Definitions and complete objective contract

Service objectives define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

Declared objective contract	Complete local exercise
Observation window	Fictional W-A, 09:00–10:00 on one fictional day; supplied records assigned to that window
Eligibility	200 user-request fixtures; 3 separate internal health checks excluded by predeclared policy
Success	All supplied schema, support-or-correct-refusal and timing-contract flags pass
Target	At least 98% success over eligible requests
Count rule	A request with several failed checks counts as one failed request
Maximum failures	floor((100–target percent) × eligible /100) for integer-percent target
Actual evidence	Constructed check flags and count arithmetic; no real requests or timing measured
Report	Success/failure counts, eligible/excluded counts, target, budget and interval identity

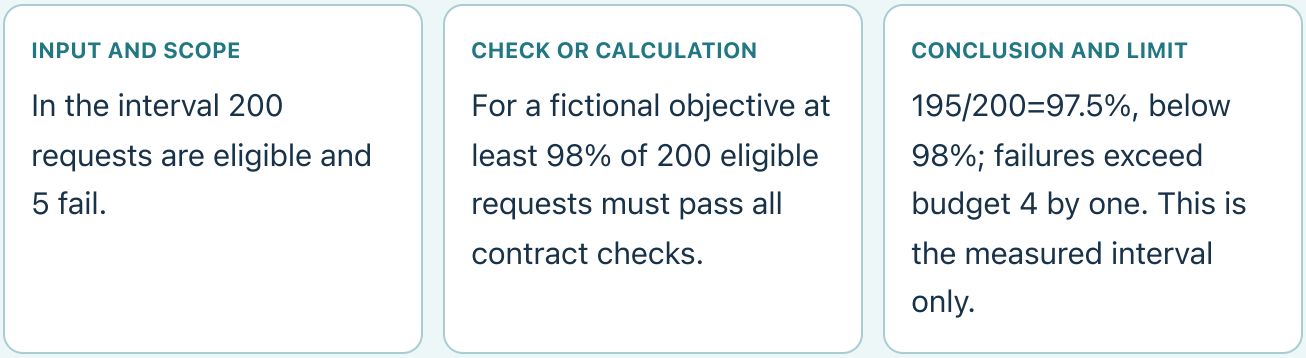
Main arithmetic and overlap-aware reasoning

Main W-A count	Calculation	Result
Eligible requests	200	Fixed denominator
Failed requests	5 distinct request IDs	Count once per request
Successes	200−5	195
Observed fixture success rate	195/200	97.5%
Maximum integer failures at 98%	(100−98)×200/100	4
Budget remaining	4−5	−1: exceeds allowance by1
Objective decision	97.5% <98%	Does not meet this window’s target
Excluded health checks	3 separate records	Report outside eligible denominator

Main195/200=97.5%, allowed 4 failures, observed 5, so misses by1. Require one failed-request count even when several checks fail. The program’s7 failed criteria belong to5 requests; a correct refusal can count as success only when it meets all declared requirements. In the picture keep original 97/3 separate from counterfactual 98/2; do not teach that a retry rewrites past reliability.

Exact integer limits floor allowed failures, so98% of12 allows none. All fixture timing flags are supplied, not real measurements.

Failure budgets and service objectives — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only. Qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

In the interval 200 requests are eligible and 5 fail.

195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Eligibility,integer and retry-history probes

Budget or evidence probe	Correct interpretation
One request fails support and schema	One failed request, two failed criteria; do not double-count the request
Correct refusal passes every required check	Can count as success under the declared contract
Three internal health checks excluded by predeclared rule	Report separately; do not remove failing eligible user requests post hoc
98% target over 12 requests	At most 0 integer failures; do not round 0.24 up to 1
No eligible requests	Empirical success rate unavailable; this helper rejects zero denominator
Aggregate target met with a severe incident	Does not prove zero critical failures or make the incident acceptable
A later retry succeeds	Retain original history and apply the predeclared retry/window rule
One constructed window meets or misses target	No guarantee about other windows or future populations

Separate pictured window and counterfactual scope

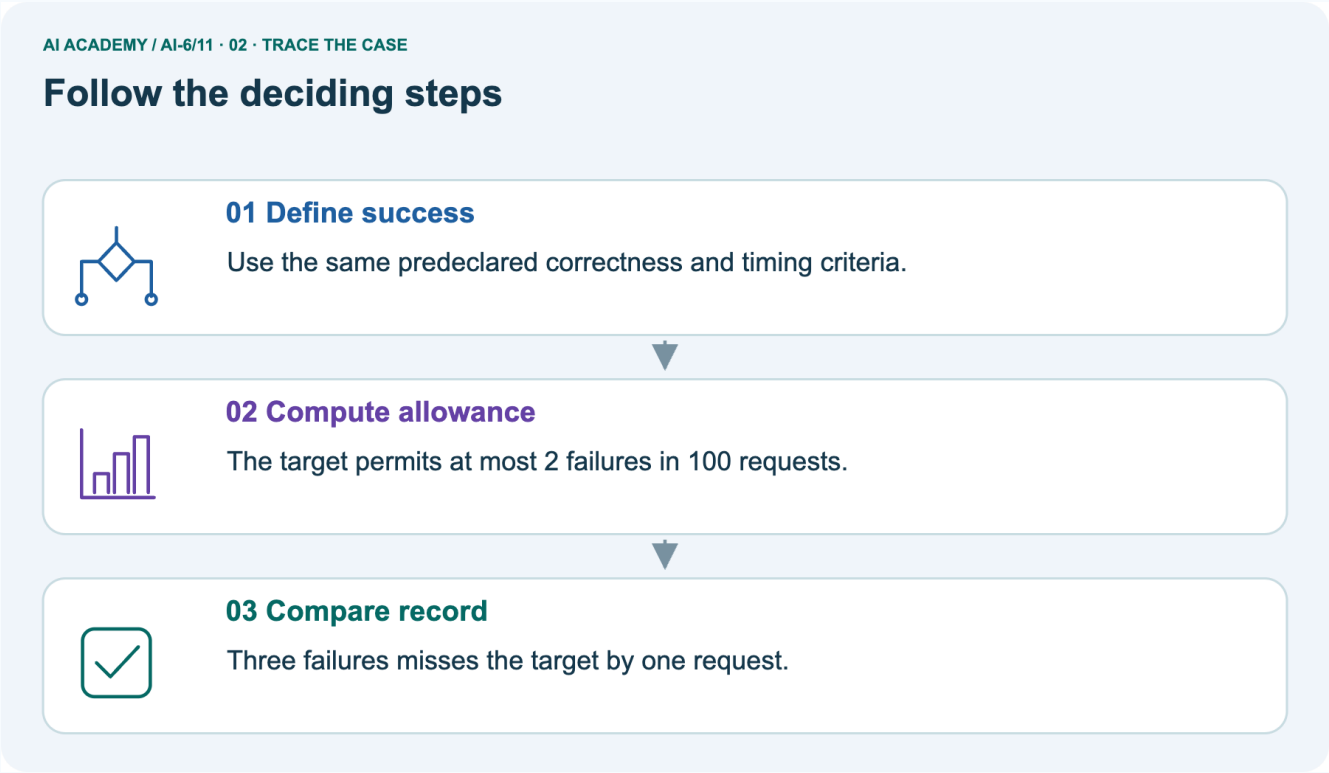
A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s.

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.

The main window has 200 eligible requests and 5 failures. The picture is a separate 100-request window:97 successes, 3 failures, target at least 98 successes. Its maximum allowance is2 failures, so it misses by1. The changed picture is a counterfactual copy with 98 successes and 2 failures, which meets the same rule. Preserve the original 97/3 history.

In a real service, a later successful retry does not automatically erase the original failed request; request identity, retry counting and window policy must be declared before measurement.

Picture trace and preserved original/copy key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define success	Use the same predeclared correctness and timing criteria.
2. Compute allowance	The target permits at most 2 failures in 100 requests.
3. Compare record	Three failures misses the target by one request.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

Change: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window.

Prediction: 98 successes and 2 failures meet the stated window target.

Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Separate pictured record · Successes / eligible · Failures · Allowed failures · Decision

> **Separate pictured record:** Original window, retain unchanged
>
> **Successes / eligible:** 97/100
>
> **Failures:** 3

> **Allowed failures:** 2
>
> **Decision:** Misses target

> **Separate pictured record:** Counterfactual copy, one different outcome
>
> **Successes / eligible:** 98/100
>
> **Failures:** 2
>
> **Allowed failures:** 2
>
> **Decision:** Meets target at equality

> **Separate pictured record:** Actual retry after an observed failure
>
> **Successes / eligible:** Requires declared request/window policy
>
> **Failures:** Do not silently erase prior failure
>
> **Allowed failures:** Keep original history
>
> **Decision:** Not supplied or measured here

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Service objective	A declared measured target over a specified scope
Failure budget	The failures allowed under that target

Term	Meaning
Eligible request	A request included under the declared denominator rule

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

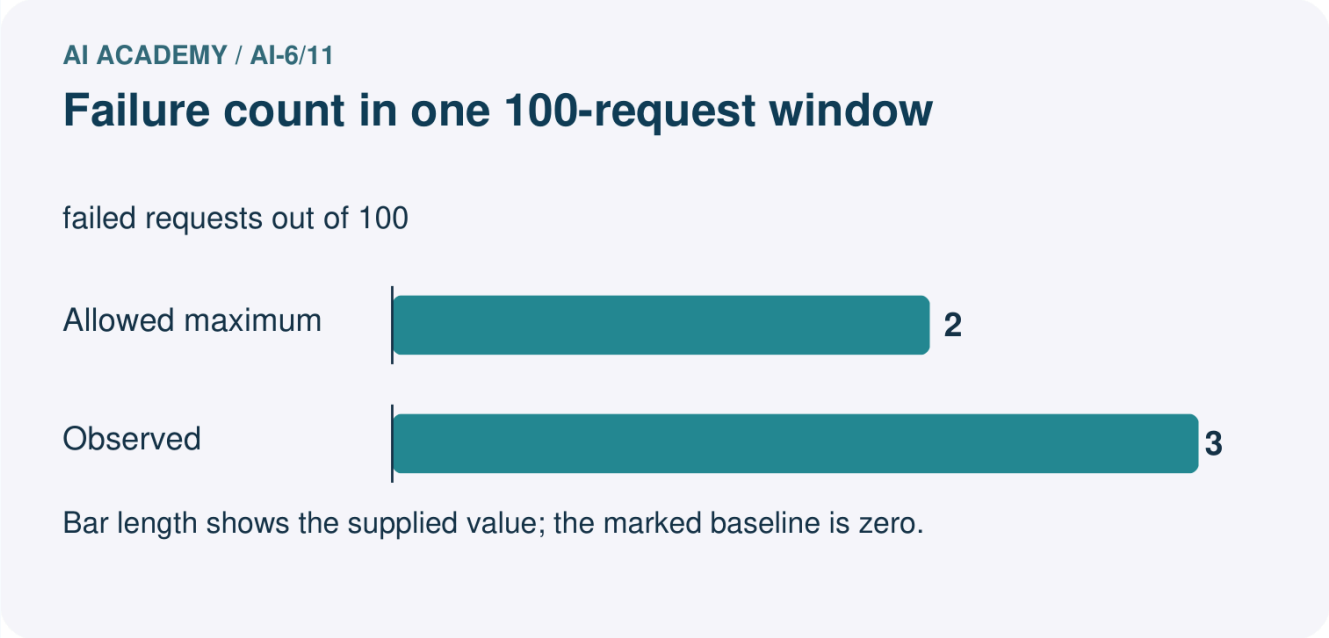
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Failure budgets and service objectives

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/11

Failure budgets and service objectives

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define success

Use the same predeclared correctness and timing criteria.



2. Compute allowance

The target permits at most 2 failures in 100 requests.



3. Compare record

Three failures misses the target by one request.

CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Meeting one interval objective does not establish every future interval or zero critical failures.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Use the same predeclared correctness and timing criteria. The target permits at most 2 failures in 100 requests. Three failures misses the target by one request. Changed case: 98 successes and 2 failures meet the stated window target. The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Changed-case reasoning: 98 successes and 2 failures meet the stated window target. The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main98% over 200:allowed 4, observed 5, 195/200=97.5%, misses by1. Picture100:original 97/3 misses; counterfactual 98/2 meets but original preserved. Independent95% over 80 allows4 failures; 5 fail gives 75 success/80=93.75%, below 95%, exceeds budget by1. Require eligibility, window and success definition, with no future or zero-critical-failure guarantee.

Diagnose the current misconceptions

The worked result establishes every future case: Meeting one interval objective does not establish every future interval or zero critical failures.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

Teacher explanation: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Service objective**s define measured reliability over a stated interval. A **failure budget** is meaningful only after success and **eligible request**s are defined.

For a fictional objective at least 98% of 200 **eligible request**s must pass all contract checks. Allowed failures are 2%*200=4. Count every eligible failure once; report excluded requests separately.

W2 Trace the worked case

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

W3 Audit the limit

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Meeting one interval objective does not establish every future interval or zero critical failures.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Meeting one interval objective does not establish every future interval or zero critical failures.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only. Qualification: Meeting one interval objective does not establish every future interval or zero critical failures.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Objective 95% over 80 **eligible request**s. Five fail. Compute allowed failures and observed success.

Teacher explanation: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Allowed four; 75/80=93.75%, so objective fails. Unknown eligibility cannot be silently excluded. Require intermediate reasoning and the stated scope. Fresh retry: Objective 90% over 50 requests with four failures.

Expected reasoning: Allowed five; success 46/50=92%, so interval meets objective. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **failure budget**s and **service objective**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **failure budget**s and **service objective**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Logs traces and privacy. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/12, ai-6/14, ai-6/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **failure budget**s and **service objective**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 195/200=97.5%, below 98%; failures exceed budget 4 by one. This is the measured interval only.

Check every learner before discussion: Use the worked case for **Failure budget**s and **service objective**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Meeting one interval objective does not establish every future interval or zero critical failures.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Require a **prediction** and a reason before comparing. Expected result: 98 successes and 2 failures meet the stated window target. Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Failure budgets and service objectives: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

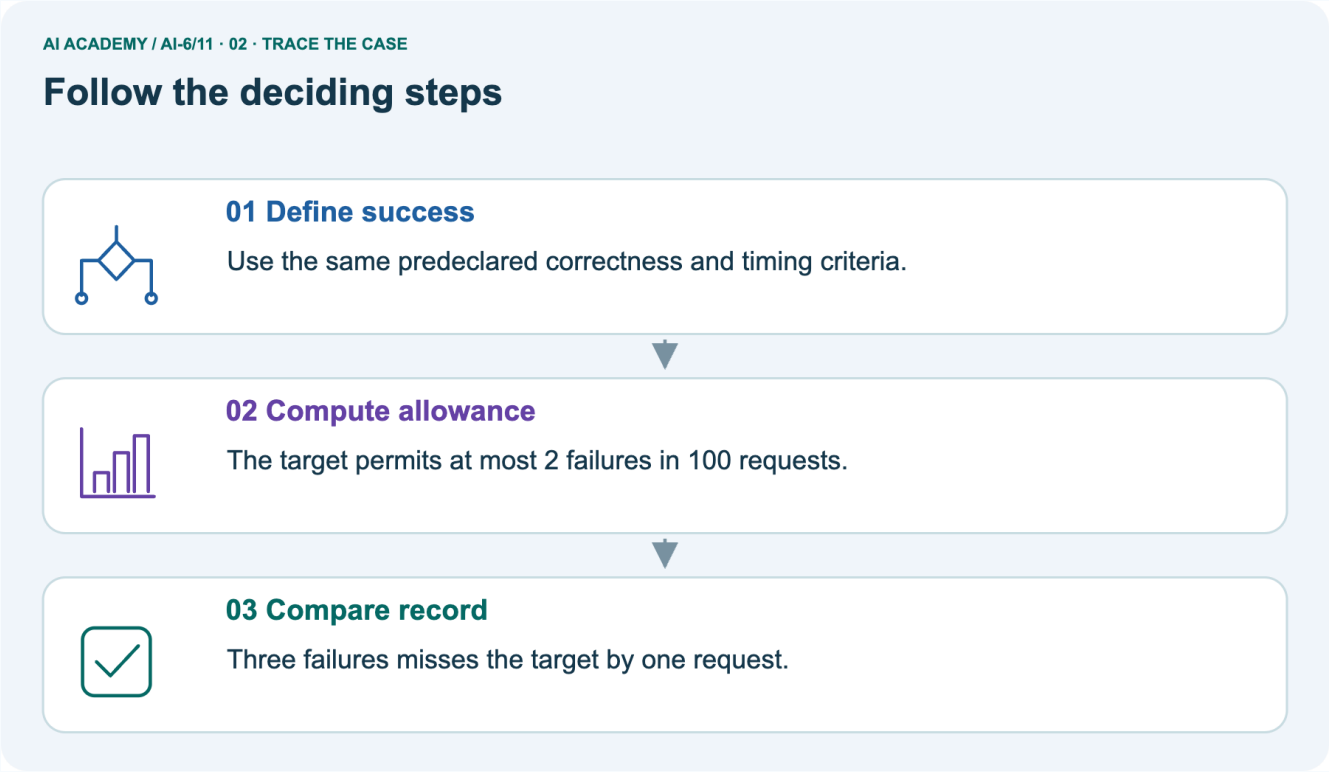
A reliability allowance. A practice service requires at least 98 successes in each window of 100 **eligible request**s. Supplied facts: Current window has 97 successes and 3 failures.

Success includes a correct refusal within the agreed time limit.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



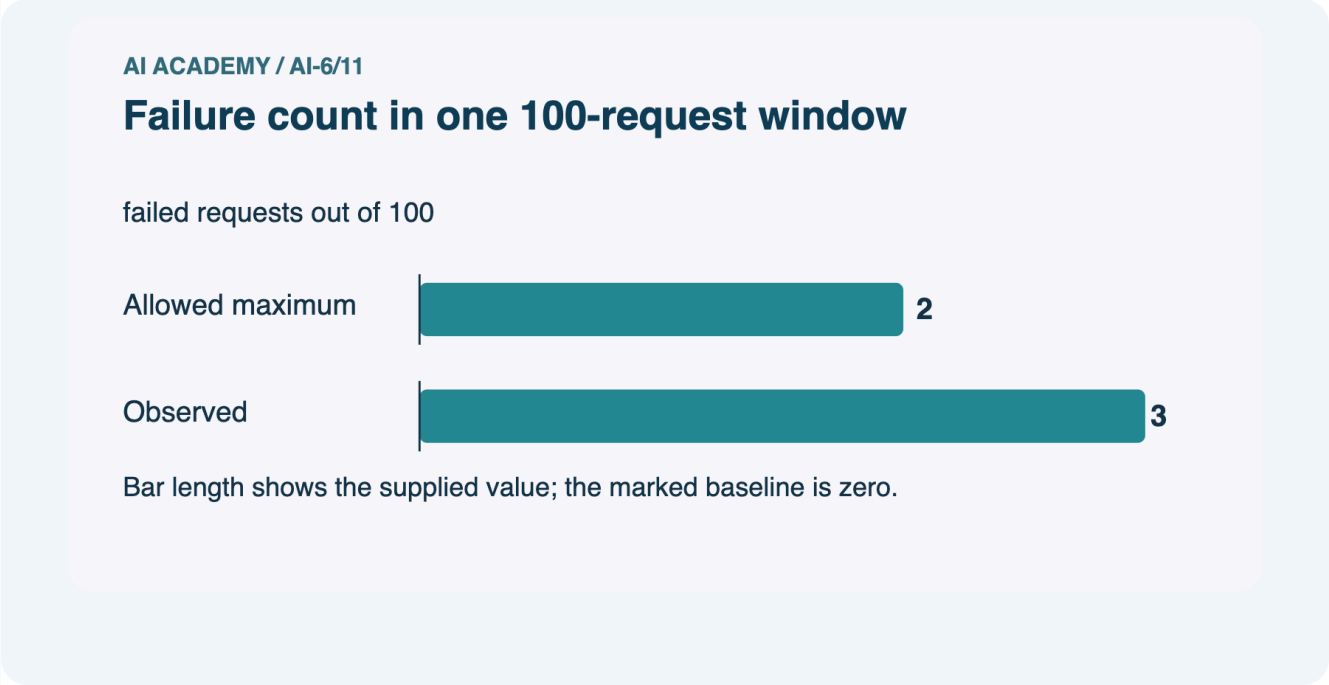
Follow the labelled picture with the matching explanation. Open the vector diagram

Define success: Use the same predeclared correctness and timing criteria. → Compute allowance: The target permits at most 2 failures in 100 requests. → Compare record: Three failures misses the target by one request.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Failure count in one 100-request window



Follow the labelled picture with the matching explanation. Open the vector diagram

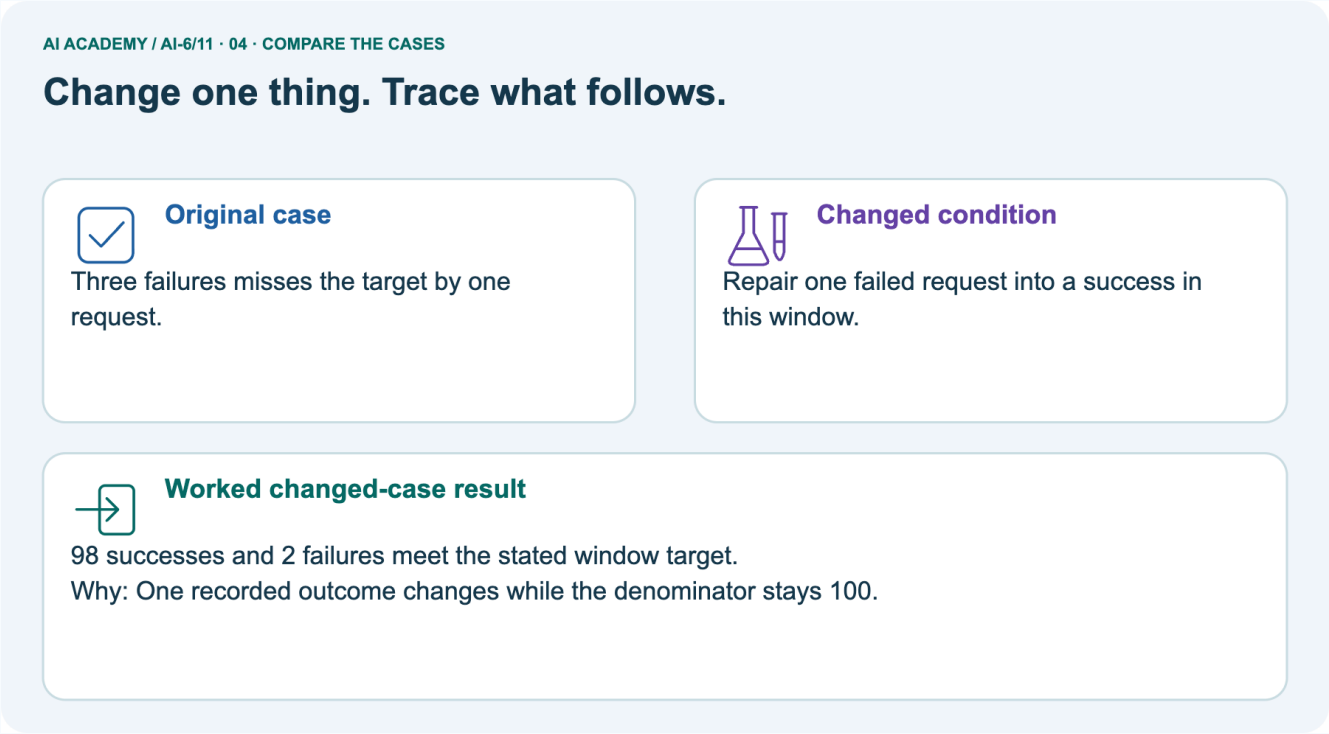
Failure count in one 100-request window. Units: failed requests out of 100. Allowed maximum: 2. Observed: 3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



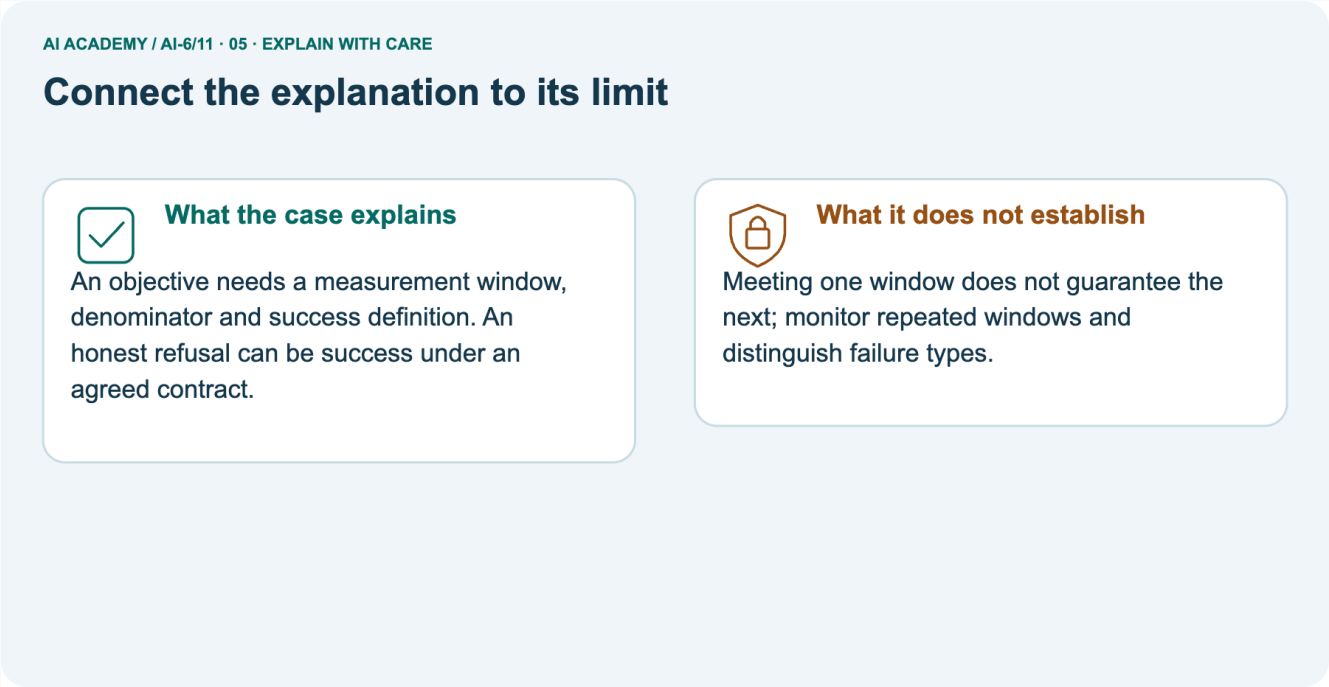
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three failures misses the target by one request. Changed condition: Counterfactual: change one failed outcome to a success in a separate copy of this 100-request window. Expected result: 98 successes and 2 failures meet the stated window target. Reason: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

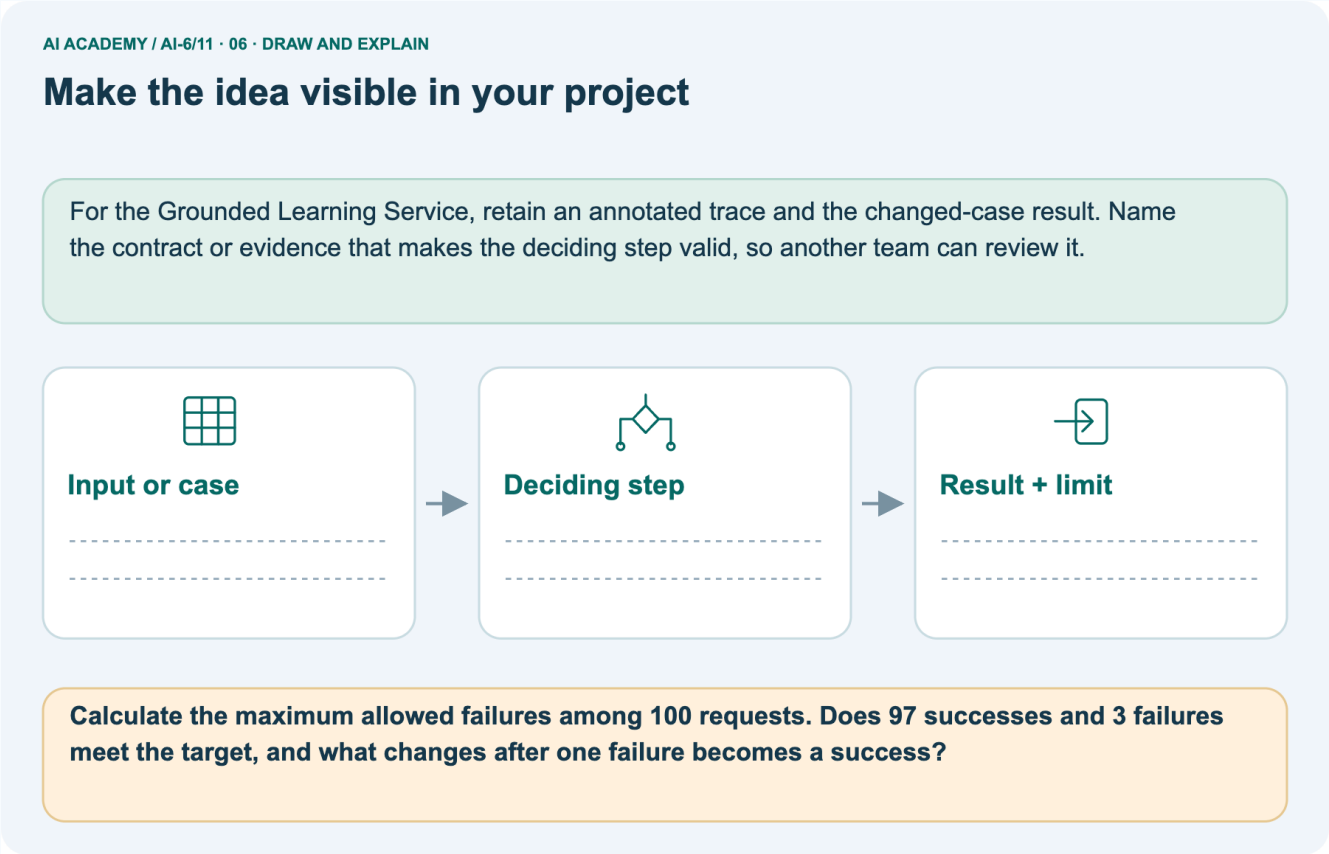
Limit: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Supported practice key: Use the same predeclared correctness and timing criteria. The target permits at most 2 failures in 100 requests. Three failures misses the

target by one request. Changed case: 98 successes and 2 failures meet the stated window target. The counterfactual copy changes one outcome while its denominator stays 100; preserve the original 97-success/3-failure window.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Failure budgets and service objectives: the case

02. Follow the deciding steps

03. Failure count in one 100-request window

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.